Telecom Customer Churn Analysis

Final Recommendations Report

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Executive Summary

A Logistic Regression model was built to predict customer churn for a telecom provider using the IBM Telco Customer Churn dataset (7,043 customers, 21 attributes). The model achieved 81.7% overall accuracy. Exploratory analysis, SQL querying, and SHAP-based explainability consistently point to **contract type** and **monthly charges** as the leading drivers of churn. Customers were further segmented into three risk tiers to support targeted retention action. This report summarizes the key findings and translates them into concrete, prioritized recommendations for the business.

26.5%	81.7%	6.2%	88.5%
Overall churn rate	Model accuracy	Customers flagged “At Risk”	Churn from month-to-month contracts

Key Findings

Area	Finding
Contract type	88.5% of churned customers (1,655 of 1,869) were on month-to-month contracts, versus only 166 on one-year and 48 on two-year contracts.
Billing	Churned customers pay noticeably higher monthly charges on average than retained customers (overall average bill: $64.76).
Model performance	81.7% accuracy overall; recall on the churn class is 57%, meaning the model still misses a portion of true churners — useful as a screening tool alongside business judgment.
Explainability (SHAP)	Confirms contract type, tenure, and monthly charges as the features with the largest influence on individual churn predictions.
Segmentation	Customers split into Loyal (4,358), Dormant (2,250), and At Risk (435, churn probability ≥ 70%) based on predicted churn probability.

Final Recommendations

The following actions are prioritized based on the size of the customer segment affected and the strength of the churn signal identified in the analysis.

1 Migrate month-to-month customers to annual plans

Offer incentives (bill credits, free add-on months, loyalty pricing) to encourage the 88.5% of churners who were on month-to-month contracts to move to one- or two-year plans, which show dramatically lower churn.

Expected impact: Highest — addresses the single strongest churn driver in the data.

2 Protect and reward the Loyal segment

Continue investing in the 4,358 “Loyal” customers (churn probability under 30%) with recognition programs and exclusive offers to reinforce their retention and defend against future risk.

Expected impact: Medium — preserves the largest and most stable revenue base.

3 Launch targeted outreach for the “At Risk” segment

Proactively contact the 435 customers with churn probability ≥ 70% with personalized discounts, retention calls, or tailored service bundles before they leave.

Expected impact: High — small segment (6.2%) but the most time-sensitive and highest-probability churners.

4 Review pricing for high-bill customers

Monitor customers with above-average monthly charges and proactively offer better-value plans or bundle discounts, since higher bills are strongly associated with churn.

Expected impact: Medium — reduces price-driven attrition across the base.

5 Strengthen customer support quality

Invest in faster resolution times and proactive service check-ins to reduce churn driven by service dissatisfaction, particularly for customers in the “Dormant” (2,250) mid-risk segment.

Expected impact: Medium-term — improves retention across all segments over time.

Conclusion

This project successfully predicted customer churn using a Logistic Regression model and combined it with SQL analysis, exploratory data analysis, SHAP explainability, and customer segmentation to build a complete picture of churn behavior. Contract type and monthly charges emerge as the clearest levers available to the business. Acting on the prioritized recommendations above — particularly contract migration and targeted outreach to At-Risk customers — offers the most direct path to reducing churn and protecting revenue.

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